

When Do Quantum Models Help? A Cross-Domain Empirical Study

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Abstract—While hybrid quantum-classical neural networks theoretically offer rich representational capacity in high-dimensional Hilbert spaces, empirical benchmarks frequently fail to isolate the quantum contribution from classical artifacting. Consistent with recent literature, we find that quantum models do not systematically outperform classical baselines across all regimes. To identify the specific conditions under which quantum models provide measurable utility, this paper introduces a strict 1-to-1 parameter-ablated benchmark across a 2×2 empirical matrix of data topology (tabular vs. geometric) and volume (sufficient vs. extreme $p \gg N$ starvation), and develops QKAResQNet, a hybrid architecture coupling a classical residual highway with trainable Quantum Kernel Alignment through a learnable gating parameter. We identify four operational boundaries: (1) a geometric utility regime under data starvation, where quantum twins outperform parameter-matched classical twins on topologically complex manifolds (up to +24.48 percentage points), while remaining strictly below the exact classical RBF-SVM ceiling and thereby acting as scalable $O(N)$ approximations of $O(N^3)$ kernel methods; (2) representational parity under multi-omics feature starvation, where a 4-qubit bottleneck matches classical network capacity with tighter variance; (3) standard-regime parity, where classical models dominate strictly due to training-time efficiency; and (4) a classical decoherence boundary, where hybrid architectures fail on quantum-native entangled data because classical compression destroys multipartite entanglement. We further validate QKAResQNet on a physical 156-qubit IBM Kingston processor, where hardware inference reaches 92.86% on real-world sensor data. These findings establish inductive bias alignment between the parameterized quantum circuit and the data manifold as a necessary condition for NISQ-era quantum utility.

Index Terms—Quantum Machine Learning, Hybrid Neural Networks, QKAResQNet, Quantum Kernel Alignment, Data Starvation, Parameterized Quantum Circuits, Inductive Bias.

I. INTRODUCTION

Parameterized quantum circuits (PQCs) have been proposed as a route to richer inductive bias and enhanced representational capacity [5]–[7]. However, the Noisy Intermediate-Scale Quantum (NISQ) era is plagued by an epistemological benchmarking crisis: a substantial fraction of papers claiming a “quantum advantage” rely on unfair comparisons, artificial datasets, or mismatched parameter counts [13]–[15]. In prac-

tice, it remains unclear exactly when hybrid quantum models genuinely provide utility over mature classical networks.

In small-data regimes, especially when the number of features greatly exceeds the number of samples ($p \gg N$), classical models suffer from interpolative collapse, while quantum models often fall victim to barren plateaus or trap-swamped optimization landscapes [10]–[12]. Generalization from limited training data remains a fundamental challenge [1]–[3].

This work focuses on a targeted empirical question: *when does the quantum component actually help?* We adopt an explore-then-develop methodology. We first explore five quantum architectures across a 2×2 empirical matrix (topology vs. volume) under strict 1-to-1 parameter ablation, and then develop QKAResQNet, an architecture that couples a classical residual highway with trainable Quantum Kernel Alignment through a learnable gating parameter γ , designed to address the barren-plateau instability and static feature-collision failures exposed during exploration. We abandon the search for universal supremacy and instead map the true operational boundaries of hybrid NISQ-era models, validating the final architecture on physical quantum hardware.

II. METHODOLOGY

A. Architectural Design

We explore three hybrid topologies: **ResQNet**, **SplitAttentionQNN**, and the hero architecture **QKAResQNet**. ResQNet combines a classical compression stage with a small 4-qubit quantum bottleneck and a residual bypass connection, building upon recent lean hybrid topologies [9], [16]. The classical front-end maps the input into a low-dimensional latent space, while the quantum block processes that representation using a hardware-efficient parameterized quantum circuit. The residual path preserves gradient flow, bypassing barren plateaus. For high-dimensional topological inputs, SplitAttentionQNN spatially chunks the feature space, passing independent representations through parallel quantum embeddings combined by a learned attention layer.

QKAResQNet retains the residual highway and dynamic gating of ResQNet but replaces the static feature map with a trainable Quantum Kernel Alignment (QKA) layer, allowing the embedding geometry to adapt to each dataset during optimization. All three hybrids employ a learnable scalar gate γ that scales the quantum branch relative to the classical highway. The forward pass of QKAResQNet is:

$$\mathbf{y} = H(\mathbf{x}) + \gamma \cdot E(Q_{\text{aligned}}(\tanh(C(\mathbf{x})))) \quad (1)$$

where H is the classical highway, C the classical compressor mapping inputs to the 4-qubit bottleneck, Q_{aligned} the trainable aligned quantum circuit, and E the classical expander. Initialized near zero, γ keeps the model classical early in training and increases as the classical path converges, progressively blending in the quantum contribution while maintaining stable gradients. Fig. 1 shows the data flow.

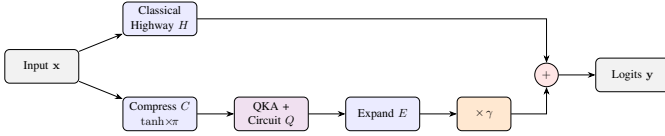


Fig. 1. The QKAResQNet architecture. A classical highway H runs in parallel with a quantum residual branch (compressor, trainable Quantum Kernel Alignment circuit, expander). The learnable gate γ scales the quantum branch before it is summed with the highway, yielding stable gradient flow and adaptive embedding geometry.

B. Unified 1-to-1 Ablation Strategy

To eliminate classical simulation artifacts, our experimental setup enforces a strict 1-to-1 replacement strategy. The classical and hybrid model twins share identical classical ingestion layers, scaling mechanisms, optimizers, learning rates, and total parameter counts. The sole variable altered is the central bottleneck: transitioning from a standard Euclidean dense layer to a unitary quantum circuit.

C. Statistical Protocol and Hardware Simulation

To accurately reflect physical NISQ constraints, the six core datasets were trained with density-matrix simulations incorporating the empirically derived single-qubit, two-qubit, and readout errors of the 133-qubit IBM Torino processor (Heron r1, 3.03% median readout error). Recognizing the extreme statistical instability of $p \gg N$ starvation regimes, model performance was evaluated across 20-seed randomized robustness sweeps, with statistical significance assessed using both Welch’s t -test and the Mann-Whitney U test to isolate genuine topological effects from initialization noise. For real-world validation, QKAResQNet was additionally deployed for inference on the physical 156-qubit IBM Kingston processor (Heron r2, 340K CLOPS).

III. EXPERIMENTS AND RESULTS: FOUR OPERATIONAL BOUNDARIES

The empirical matrix spans geometric (Swiss Roll, Circles, Moons), chemical (QSAR), tabular (Breast Cancer),

multi-omics (Leukemia, Colon), high-dimensional imaging (Brain RNA-seq, Indian Pines), and quantum-native (NTangled) datasets, with real-world ESP32 accelerometer gesture data used for physical-hardware validation. Table I summarizes the hero QKAResQNet pair against its parameter-matched CKAResCNet twin across all eleven datasets; the per-regime analyses follow.

TABLE I
QKAResQNet vs. PARAMETER-MATCHED CLASSICAL TWIN (CKAResCNet) ACROSS ALL DATASETS (20-SEED MEAN ACCURACY, %)

Dataset	Classical	Quantum	Δ / Verdict
Swiss Roll	67.73	82.16	+14.42 Q wins
Circles	65.61	76.50	+10.89 Q wins
Moons	86.17	91.26	+5.09 Q wins
QSAR	82.21	82.61	+0.40 parity
Breast Cancer	88.50	85.14	parity
Leukemia	70.85	69.55	parity ($p=0.70$)
Colon	66.26	63.35	parity
Brain RNA-seq	43.61	43.11	parity ($p=0.93$)
Indian Pines	61.61	61.94	parity
NTangled	49.83	49.67	both fail (~ 50)

A. The Topological Advantage (Geometric Starvation)

To evaluate inductive bias on complex continuous geometries under data starvation, models were tested on the Swiss Roll manifold with a 90/10 train-test split across 20 random seeds.

TABLE II
SWISS ROLL (20-SEED): GEOMETRIC UTILITY UNDER STARVATION

Architecture Family	Classical Twin	Hybrid Quantum Twin
Split Attention	67.29% ($\pm 11.8\%$)	91.77% ($\pm 4.4\%$)
MLP Base	75.32% ($\pm 12.0\%$)	84.44% ($\pm 9.3\%$)
Residual (ResNet/ResQNet)	69.93% ($\pm 14.2\%$)	82.56% ($\pm 7.8\%$)
QKA Residual (CKA/QKAResQNet)	67.73% ($\pm 11.8\%$)	82.16% ($\pm 9.2\%$)
RBF-SVM (kernel ceiling)	98.89%	

As shown in Table II, the quantum twins demonstrate a consistent geometric utility under starvation. The *SplitAttentionQNN* reaches 91.77% accuracy, exceeding its classical twin by +24.48 percentage points with tighter variance, while the hero *QKAResQNet* contributes +14.42pp over its CKAResCNet twin ($p < 0.001$). The pattern holds across the other two synthetic manifolds: QKAResQNet beats its twin by +10.89pp on Circles and +5.09pp on Moons. The phase-encoded Hilbert space provides a stronger inductive bias for unrolling non-linear manifolds than flat Euclidean space. Crucially, none of the hybrids exceed the exact RBF-SVM ceiling of 98.89%, confirming that they function as scalable $O(N)$ approximations of an $O(N^3)$ kernel method rather than surpassing it; exact quantum kernels themselves face exponential concentration at scale [8].

A complementary finding is the failure of the *static* Quantum SVM on these same continuous manifolds (Circles: 59.27%; Moons: 63.92%), caused by periodic feature collisions in the fixed ZZ-feature map. Trainable kernel alignment

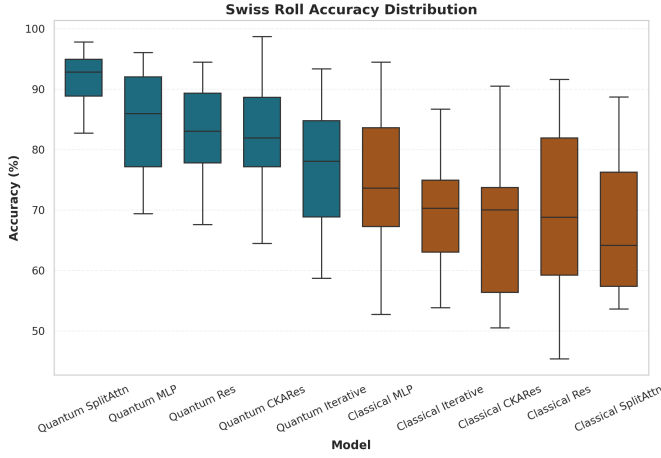


Fig. 2. Geometric utility on the Swiss Roll manifold (20-seed mean $\pm$ std). Every quantum twin outperforms its parameter-matched classical twin; the SplitAttentionQNN attains the largest margin (+24.48pp). All hybrids remain below the exact RBF-SVM ceiling of 98.89%, consistent with the kernel-approximation interpretation.

in QKAResQNet recovers performance, whereas a static encoding cannot, isolating adaptive embedding as the mechanism of recovery.

B. Representational Parity ($p \gg N$ Starvation)

To test stability in a severe feature-starvation regime, Leukemia (7,129 features) and Colon (2,000 features) were evaluated with microscopic sample sizes over 20 random seeds.

TABLE III
EXTREME STARVATION (20-SEED): REPRESENTATIONAL PARITY

Dataset	Model	Accuracy (%)	Total Params
Colon	Classical ResNet	63.63 $\pm$ 11.4	12,037
	Hybrid ResQNet	65.13 $\pm$ 8.7	12,029
	Hybrid QKAResQNet	63.35 $\pm$ 11.1	12,029
Leukemia	Classical ResNet	71.42 $\pm$ 11.2	42,811
	Hybrid ResQNet	68.91 $\pm$ 10.0	42,803
	Hybrid QKAResQNet	69.55 $\pm$ 9.1	42,803

Table III and Fig. 3 demonstrate representational parity. The 4-qubit quantum bottleneck matches the capacity of the massively parameterized classical bottleneck ($\sim 65\%$) using a near-identical parameter count, with the aligned hybrids reaching statistical parity (Leukemia: ResQNet $p = 0.471$, QKAResQNet $p = 0.701$; Colon: parity). The unitary constraints act as an implicit regularizer against the chaotic $p \gg N$ noise [4], yielding tighter standard deviations (Leukemia: $\pm 9.1\%$ for QKAResQNet vs $\pm 11.2\%$ classical).

Architecture selection is decisive in this regime. Whereas the aligned residual hybrids reach parity, the locality-biased *SplitAttentionQNN* causes statistically significant *harm* on the same genomic data (Leukemia -6.40pp , $p = 0.0016$; Colon -10.24pp , $p = 0.0098$), because chunking destroys the non-local feature correlations that carry the biological signal.

Inductive bias misalignment therefore degrades performance rather than merely failing to help, establishing alignment as a necessary condition for utility.

This pattern extends to two further high-dimensional datasets. On Brain RNA-seq (7 classes, $p = 23,433$, $N = 100$), every architecture is pinned near the imbalanced multi-class floor; the aligned QKAResQNet reaches exact parity (43.11% vs 43.61%, $p = 0.93$) while SplitAttentionQNN collapses to 15.33%, the most extreme manifestation of the misalignment failure. On the Indian Pines hyperspectral subset ($p = 200$), all pairs reach parity and the Quantum SVM nearly matches the RBF-SVM ceiling (71.97% vs 72.28%), because the spectrally smooth features do not trigger the periodic collisions that break static kernels on geometric manifolds.

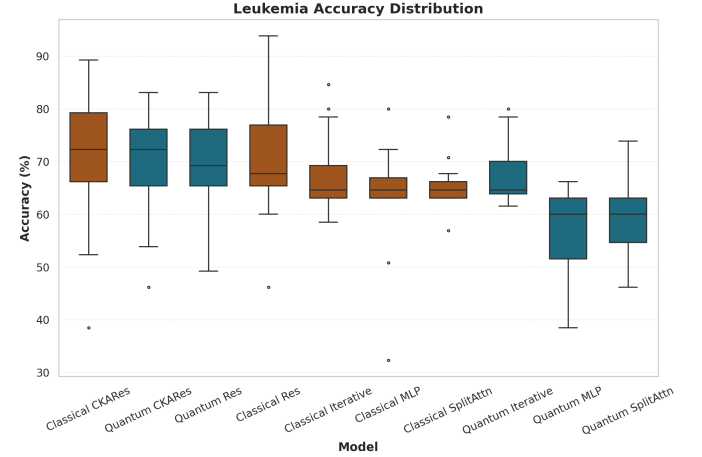


Fig. 3. Leukemia 19-seed accuracy by architecture pair (mean $\pm$ std). Accuracies sit near a generalization ceiling of $\sim 65\%$ due to the microscopic training set. The aligned quantum twins (ResQNet, QKAResQNet) reach statistical parity with tighter variance, while the misaligned SplitAttentionQNN underperforms.

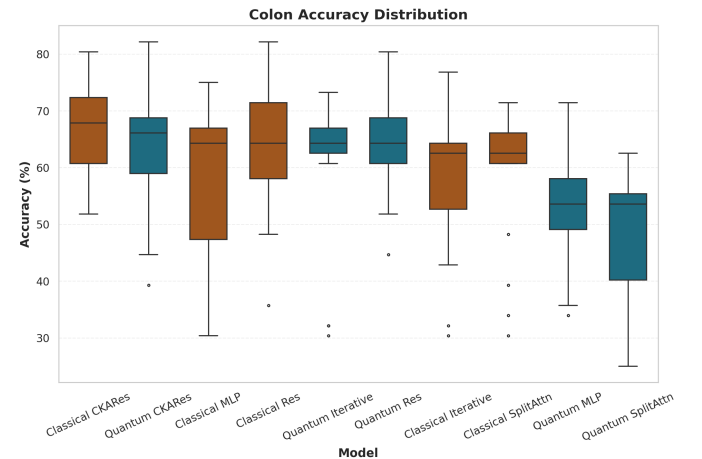


Fig. 4. Colon cancer gene expression 19-seed accuracy by architecture pair (mean $\pm$ std). The high-dimensional ($p = 2,000$) starvation regime reproduces the Leukemia pattern across a second genomic domain: aligned quantum twins reach parity with their parameter-matched classical twins, supporting the generalizability of the representational-parity boundary.

C. Standard Parity & The Execution Time Penalty

On standard tabular datasets with sufficient sample density (Breast Cancer, QSAR), the hybrid models tied the classical baselines (e.g., Breast Cancer ResNet 89.72% vs ResQNet 88.79%). However, this parity exposes a severe practical bottleneck. Classical total training times averaged ~ 0.01 seconds, whereas the quantum models required 24 to 419 seconds over the full 20-epoch run, a three- to four-order-of-magnitude penalty. In the NISQ era, any geometric utility trades off directly against severe wall-clock time penalties.



Fig. 5. The execution-time bottleneck. Total training time (20 epochs) across evaluated datasets on a logarithmic scale. While standard tabular datasets exhibit representational parity, quantum-circuit simulation imposes a severe wall-clock penalty (milliseconds vs. tens to hundreds of seconds).

D. The Classical Decoherence Boundary

To evaluate hybrid performance on native quantum states, the architectures were tasked with classifying the multipartite entangled NTangled dataset.

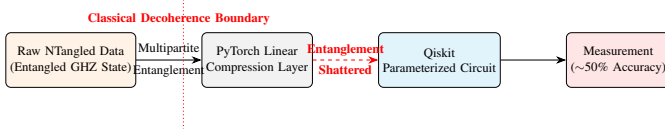


Fig. 6. The Classical Decoherence Boundary. The classical dimensionality reduction layer acts as an artificial measurement, irreversibly destroying non-local entanglement before it can be processed by the quantum circuit.

While purely quantum literature reports $>90\%$ accuracy on this dataset, every single hybrid architecture evaluated stalled at precisely 50% (random guessing). As illustrated in Fig. 6, the necessary PyTorch classical dimensionality reduction layer mathematically acts as an artificial measurement [18]. This observation is consistent with the structural properties of entangled quantum-native datasets [17], which encode non-classical correlations that classical compression irrevocably destroys. This empirically defines the *Classical Decoherence Boundary*: standard Classical-to-Quantum pipelines are fundamentally incapable of processing native quantum entanglement.

E. Real-World Validation Across Hardware Conditions

To move beyond synthetic manifolds and simulation, QKAResQNet was evaluated on real-world ESP32 accelerometer gesture data (10 classes, complex multi-sensor geometry) under three conditions (Table IV). Under noiseless simulation, QKAResQNet exceeds its classical CKAResCNet twin by $+1.81\text{pp}$ (89.39% vs 87.58%) with lower variance, consistent with the synthetic-manifold results. Under the simulated IBM Torino profile (Heron r1, 3.03% readout error), the relationship reverses (-1.82pp), quantifying the readout-dominated noise gap of the older device. On the physical 156-qubit IBM Kingston processor (Heron r2), hardware inference reaches 92.86%, matching the RBF-SVM and exceeding both the classical twin (85.71%) and its own noisy-simulator inference (85.71%); that the device beats its calibrated simulator indicates the noise model is more pessimistic than the hardware's behaviour during the run.

TABLE IV
GESTURE CLASSIFICATION ACROSS NOISE AND HARDWARE CONDITIONS

Condition	Classical	Quantum
Noiseless simulation	87.58 (± 7.1)	89.39 (± 6.5)
IBM Torino noise	89.55 (± 9.9)	87.73 (± 10.3)
IBM Kingston (hardware)	85.71	92.86
RBF-SVM (ceiling)	92.86 / 91.21	

The γ -gated residual design is the mechanism behind this hardware resilience: the classical highway carries the bulk of the prediction, so transient quantum errors degrade the output gracefully. The evaluation batch required 84 seconds of wall-clock time, dominated by CPU-QPU input/output latency rather than gate execution, identifying classical-quantum communication overhead as the principal engineering bottleneck for present-day deployment. As a directional real-world signal (5-fold CV, $N = 56$), this awaits larger labeled datasets for statistical confirmation.

IV. CONCLUSION

This study identifies four operational boundaries for hybrid neural networks. The results do not support a universal quantum advantage; instead, quantum utility is strictly bound to inductive bias alignment between the parameterized circuit and the data manifold. On complex geometries under starvation, quantum feature maps offer measurable topological utility, with QKAResQNet exceeding its classical twin by $+14.42\text{pp}$ and the best-aligned pair by $+24.48\text{pp}$, though all hybrids remain bounded by the exact RBF-SVM kernel ceiling. On extreme $p \gg N$ genomics, a 4-qubit bottleneck matches the representational capacity of massive classical networks with tighter variance, while a misaligned architecture (SplitAttentionQNN) causes statistically significant harm. These benefits are constrained by severe NISQ execution-time penalties and by the classical decoherence of native quantum states. Physical deployment on IBM Kingston confirms that QKAResQNet retains its accuracy on real hardware (92.86%), with classical-quantum I/O latency, not circuit execution,

as the dominant practical barrier. Future work will expand physical-hardware deployment and incorporate larger real-world geometric datasets to convert these directional results into statistically confirmed claims.

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